

2021.

Q2.

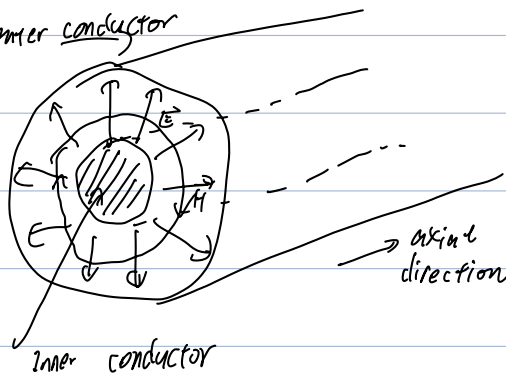
(a). potential source of interference:

Lightning

Internal high-speed circuit

Other RF signal in the ambient environment.

(b) coaxial conductor



(1) (1) Coaxial has shielding $\Rightarrow$ more resilient to RF interference.

(2) Coaxial does not generate interference with other electronic device.

(d) high gain and efficiency . $\left\{ \begin{array}{l} \text{channel capacity.} \\ \text{cross-talk} \end{array} \right.$

(e) Central frequency $f_c = \frac{11.7 + 11.7}{2} = 11.2 \text{ GHz}$

Wave length : $\lambda_c = \frac{c}{f_c} = 26.77 \text{ mm}$

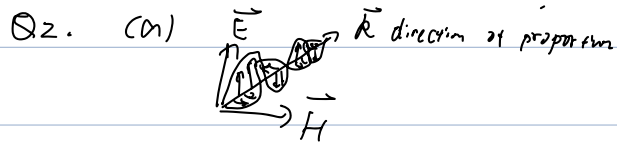
(f) $W = 2\pi f_c$

skin depth
 $\delta = \sqrt{\frac{2}{\omega \mu \sigma}} = \sqrt{\frac{1}{\pi f_c \mu \sigma}} = 564.39 \text{ nm}$

$d \geq 3\delta \approx 1.69 \mu\text{m}$

$$Q1) \quad v = \frac{c}{\sqrt{\epsilon_r}} = 1.02 \times 10^8 \text{ m/s}$$

$$\beta = \omega \sqrt{\epsilon \mu} = 2\pi f_c \cdot \sqrt{\epsilon_r \epsilon_0 \mu_0} = 688.38 \text{ m}^{-1}$$



TEM: Electric field and magnetic field are on the same plane and orthogonal to the wave propagation direction.

Method of coupling:

(1) Capacitive

(2) Inductive

(3) Conduction

(4) Radiation

$$(b). \quad \text{i.e.} \quad \begin{cases} \vec{\nabla} \cdot \vec{D} = 0 \\ \vec{\nabla} \cdot \vec{H} = 0 \end{cases} \Rightarrow \begin{cases} \vec{\nabla} \cdot \vec{D} = \frac{\partial}{\partial x} (D_m \cos(\omega t - \beta z)) = 0 \\ \vec{\nabla} \cdot \vec{H} = \frac{\partial}{\partial y} (H_m \cos(\omega t - \beta z)) = 0 \end{cases}$$

$$\vec{D} \cdot \vec{B} = \vec{D} \cdot (\mu \vec{H})$$

Q3. (a). $2 \sim 3$ times the skin depth.

(b). longer the r_s , higher the Q_s

$$a. \quad Z = V/I, \quad V = \int E dl, \quad I = \int H dl$$

A numerical method e.g. finite element method (FEM)

is used to solve the E and H fields in space.

(d) (i) Reactive coupling from surrounding high-speed circuits.

(ii) lightning

(iii) Other RF signal in the ambient env.

Q4. (a)
$$T = \frac{Z_W - Z_{Air}}{Z_W + Z_{Air}} \quad T = \frac{2Z_{in}}{Z_W + Z_{Air}}$$

(b) (i)
$$Z_{in} = \sqrt{\frac{j\omega L}{8 + j\omega L}} = \sqrt{\frac{j \cdot 2\pi \cdot 10^3}{8 + j \cdot 2\pi \cdot 10^3}} = \sqrt{\frac{3.508 \times 10^{-3} + 2.230 \times 10^{-4}j}{8 + j \cdot 2\pi \cdot 10^3}} = \sqrt{2.230 \times 10^{-4} \angle 90^\circ} = 0.0149 \angle 45^\circ$$

$$Z_{Air} = \sqrt{\frac{\mu}{\epsilon}} = 376.73$$

$$T = \frac{Z_{in} - Z_{Air}}{Z_{in} + Z_{Air}} = \frac{0.0149 \angle 45^\circ - 376.73}{0.0149 \angle 45^\circ + 376.73} \approx -1$$

(ii)
$$\delta = \sqrt{\frac{2}{\omega \mu \epsilon}} = 2.675 \text{ mNp} \Rightarrow \alpha = \frac{2}{\delta}$$

$$20 \log_{10} e^{-\frac{2}{\delta}} = -60 \Rightarrow \frac{2}{\delta} \cdot 20 \log_{10} e = 60$$

$$\Rightarrow \delta = \frac{60 \delta}{20 \log_{10} e} = 18.478 \text{ mm}$$

(c). be shielded with metal box

microwave oven door is shielded with metal grid.

Microwave can cause health hazard. due to

heating of skin. It can avoid exposure of skin

to high power microwave.

Q5. (i).
$$V_1^+ = \frac{Z_0}{Z_0 + Z_0} V_S \Rightarrow V_S = \frac{50 + 75}{75} \cdot 6 = 10 \text{ V}$$

(ii).
$$V = \frac{c}{\sqrt{\epsilon_r}} = 1.999 \times 10^8 \text{ m/s}$$

If there is no fault:
$$\frac{T_S}{T_L} = \frac{50 - 75}{50 + 75} = -0.2$$

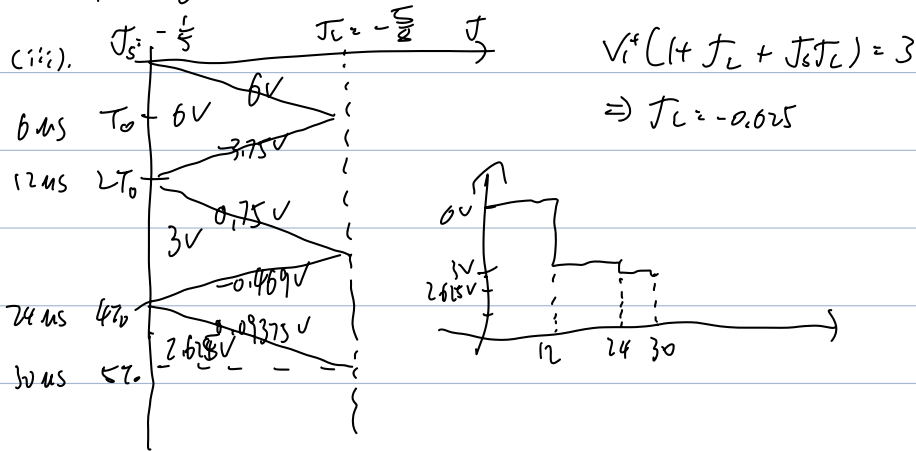
$$V_1^- = T_L \cdot V_1^+ = 6$$

$$\Rightarrow V_0(T_2) = 6 + 6 - 1.2 = 10.8 \text{ V} \neq 3 \text{ V}$$

$$V_1^+ = T_S V_1^- = -1.2$$

so there should be one fault.

$$2T_0 = 2\frac{L}{v} \Rightarrow t = vT_0 = 1200 \text{ m}$$



(b). 50 50 100

$$J_0 = \frac{50-50}{50+50} = 0 \quad J_L = \frac{100-50}{100+50} = \frac{1}{3} = 0.333$$

$$V_i^* = \frac{50}{50+50} \times 10 = 5 \text{ V}$$

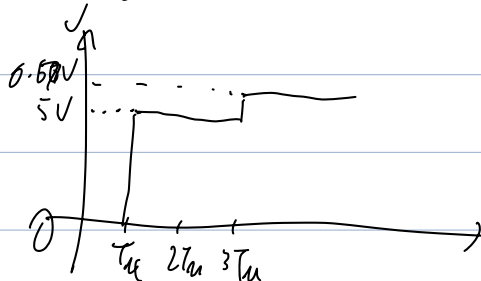
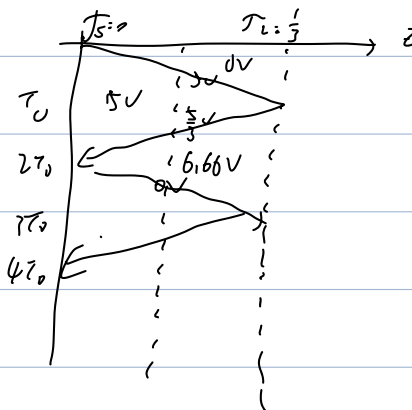
$$V_i = 5 \times \frac{1}{3} = \frac{5}{3}$$

$$V_L^* = 0$$

$$v = \frac{L}{J_0 T_0} = 1.499 \times 10^8 \text{ m/s}$$

$$T_0 = \frac{1}{v} = 6.671 \text{ ns}$$

$$T_u = \frac{T_0}{2} = 3.335 \text{ ns}$$



$$Qb. (a). P_r = P_t \cdot G_r G_e \cdot \left(\frac{\lambda}{4\pi r}\right)^2$$

$$G_e = 13.5 \text{ dB} = 22.39$$

$$G_r = 5.3 \text{ dB} = 3.39$$

$$\lambda = \frac{c}{f} = 0.19 \text{ m}$$

$$\Rightarrow P_r = 1.063 \text{ nW} = -59.73 \text{ dBm}$$